Li Liu

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Spatial reasoning · Human-agent Collaboration

Research Projects

Spatial Intelligence in Foundation Models (COLM 2026, NeurIPS 2024, one paper under review): Led a three-paper line of work on spatial understanding in frontier models: diagnosing failures, improving models through training, and testing a grounded agent with users.

- **Diagnose:** Identified two systematic failure modes of frontier models: they answer when the image cannot support an answer, and they read spatial layouts but cannot construct them from text (the strongest model places only about half of relations correctly). Released benchmarks for both (Directional Guidance, Text2Space).
- **Improve:** Fine-tuned small open VLMs to outperform frontier proprietary models at guiding users to capture missing visual information, using automatically generated training data. Introduced layout drawing as an auxiliary training task for LLMs, significantly improving text-only spatial reasoning and generalizing to external benchmarks.
- **Application:** Built an end-to-end, plug-and-play accessible app that turns any outdoor location into an audio-haptic map on an everyday laptop. Improved a speech-native model's spatial accuracy through context engineering anchored to the user's point of touch. In a study with 14 blind and low-vision participants, users read maps on their own and found discovery seeds, places they would not have known to query, extending access beyond query-driven search.

Trustworthy Multimodal AI for Healthcare (IEEE TIM, MICCAI, EJR Open): Built uncertainty-aware models for clinical and sensor data that quantify the reliability of each prediction.

- Proposed CPSC, a low-cost conformal predictor with distribution-free coverage guarantees. [pdf]
- Contributed to SwinMM, masked multi-view learning for 3D medical image segmentation (MICCAI 2023). [pdf]
- Co-developed a credibility-based COVID-19 risk-stratification system that stays reliable under data shift (EJR Open). [pdf]

Selected Publications

[1] **S. Huang***, **L. Liu*** et al. "Learning to Draw ASCII Improves Spatial Reasoning in Language Models." *COLM* 2026. (*co-first author) [pdf]

[2] **L. Liu** et al. "Right This Way: Can VLMs Guide Us to See More to Answer Questions?" *NeurIPS* 2024. [pdf]

[3] **L. Liu** et al. "TouchingSpace: An Audio-Haptic Map for Blind and Low-Vision Readers." *Under review*, 2026.

[4] **L. Liu** et al. "Pattern Discovery with Wide-Lens Analysis and Sharp-Focus Validation." *WWW Demo Track* 2026. [pdf]

[5] **L. Liu** et al. "CPSC: Conformal Prediction With Shrunken Centroids for Efficient Prediction Reliability Quantification and Data Augmentation." *IEEE Trans. Instrumentation and Measurement* 2022. [pdf]

Experience

Amazon — Seattle, WA
Applied Scientist Intern — *June 2026 – Sept. 2026*

- **Typed Meta-Skills: Structured Skill Engineering for Agents**: Encoded free-text SKILL.md prompts as typed, versioned graphs of each skill's solution approach and key steps (graph engineering), serving as documentation for people and executable instructions for agents. Cut agent output tokens by about 50% at equal or better task performance.
- **Pipeline-Wide Failure Attribution**: Linked each skill to its evidence and rollouts so each failure traces to the agent, the task, or the benchmark, surfacing benchmark defects.

Amazon — Santa Clara, CA
Applied Scientist Intern — *June 2025 – Sept. 2025*

- **Taxonomy Discovery from Unlabeled Conversations (WWW 2026)**: Developed a human-in-the-loop framework that grows a validated taxonomy over large unlabeled conversation logs from a few seed keywords, with every pattern traceable to its source. Domain-agnostic and relevant to LLM error analysis and usage analytics; demonstrated on 5,000 customer-service conversations.

Amazon — Palo Alto, CA
Applied Scientist Intern — *June 2024 – Sept. 2024*

- **Evidence-Grounded Recommendation**: Built a retrieval-augmented recommender for subjective product queries that cites the user-generated content supporting each suggestion.

Nissan Research Center — Santa Clara, CA
Data Scientist Intern — *July 2023 – Sept. 2023*

- **Driver Engagement in Autonomous Driving**: Developed a time-series model of driver engagement from real-world driving logs, improving prediction accuracy by 16%.

Education

UC Santa Cruz, Ph.D. in Computer Science and Engineering, advisor: Leilani Gilpin — 2022 – 2027 (est.)

Zhejiang University, M.Eng. in Control Science and Engineering — 2019 – 2022

Academic Activities

Reviewer: CVPR, ECCV, ICLR, NeurIPS, CIKM, WACV. **TA:** Python, Computer Vision. **Mentor:** Foothill SLI, UCSC SIP, Stanford summer research.